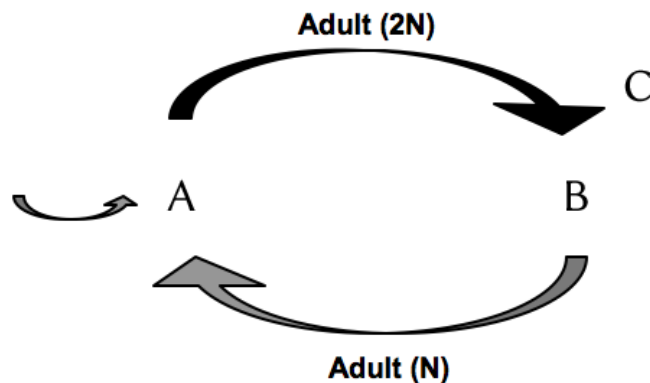


Why do protists make poor mathematicians? Each question is worth 2 points except where noted. Good luck!

Circle True or False. Read each statement completely before answering.

1. True **FALSE** Fungi and animals both spend most of their life cycle in the diploid stage.
2. **TRUE** False Adhesion is the hydrogen bonding of water molecules to the cell walls of the xylem.
3. True **FALSE** High humidity increases the rate of transpiration.
4. True **FALSE** Plants spend most of their energy needs for photosynthesis.
5. True **FALSE** A chemoheterotrophic protist is more likely to have an eyespot than a photoautotrophic protist.
6. **TRUE** False Open circulatory systems are less efficient than closed circulatory systems.
7. **TRUE** False Dynein is the motor protein found in eukaryotic flagella.
8. **TRUE** False Without H^+ ions bacteria could not move their flagella.
9. True **FALSE** Plants seedlings display negative gravitaxis by growing away from gravity.
10. True **FALSE** A signal is not received at the organismal level in protists or bacteria.

Use the following figure to fill in the blanks of the next four questions.



11. The process (not the stage) that occurs at A is **Syngamy**
12. The stage at B (that results from C) is called **Spore**
13. The process at C is **Meiosis**
14. Which of the three (A, B or C) does not exist in the animal lifecycle. **B**

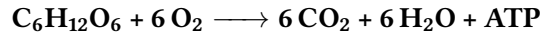
Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Active Diffusion	Gametophyte	Pseudopodia
Amplification	Gastrovascular Cavity	Reception
Binary Fission	Haplogamy	Response
Chemotaxis	Karyogamy	Run and Tumble
Cytokinesis	Magnification	Second Messenger
Cytoplasmic Streaming	Mechanotaxis	Signalation
Cytotaxis	Meiosis	Sporophyte
Diplogamy	Mitosis	Syngamy
Dive and Bomb	Netflix and Chill	Thigmotaxis
Drive and Orient	Passive Diffusion	Touchyfeelytaxis
Galvanotaxis	Plasmosyngamy	Transduction

- | | |
|----------------------------------|--|
| 15. Gastrovascular Cavity | Responsible for digestion and nutrient distribution in flatworms. |
| 16. Second Messenger | A receptor activates these ions inside a cell. |
| 17. Transduction | A signal is transmitted to a target area inside a cell. |
| 18. Plasmosyngamy | Life cycle stage where fungal cells have fused but nuclei have not. |
| 19. Cytokinesis | Division of the cell after mitosis. |
| 20. Thigmotaxis | Locomotion in response to touch. |
| 21. Cytoplasmic Streaming | Transports nutrients throughout a cell. |
| 22. Amplification | Increases the strength of a weak signal. |
| 23. Gametophyte | Mosses spend most of their life in this life cycle stage. |
| 24. Run and Tumble | Allows bacteria to move towards a chemical cue. |
| 25. ANY ANSWER | Name the band that you think is Taylor's favorite (not in list above). |

Keep on keepin' on!

26. (4 points) You knew it was coming...Write a *balanced* chemical equation for cellular respiration. In addition, include the molecule that is used to accomplish work in the cell.



Subtract 1 pt if one prefix number is missing or wrong (e.g., 4 H₂O, 2 pts if more than 1 is wrong.

Subtract 1 pt if one chemical formula is wrong (e.g., C₁₂H₆O₆), 2 pts if one missing or more than 1 wrong. Subtract 2 pt if missing ATP.

Ask me if uncertain.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

27. Galvanotaxis occurs when
- A. plants respond to gravity.
 - B. protists respond to touch.
 - C. protists respond to an electrical field.**
 - D. animals respond to temperature.
 - E. plants respond to hormones.
 - F. animals respond to metal ions.
28. Most of the water taken up by a plant
- A. is converted to sugars during photosynthesis.
 - B. transports sugars in the phloem.
 - C. is lost during transpiration.**
 - D. keep guard cells turgid.
 - E. enters the central vacuoles of cells.
29. In terms of structure, cilia are most like
- A. myosin filaments.
 - B. bacterial flagella.
 - C. eukaryotic flagella.**
 - D. actin filaments.
 - E. microtubules.
30. When ATP attaches to the myosin head
- A. the actin head attaches to the myosin filament.
 - B. the myosin head attaches to the actin filament.
 - C. the myosin head releases from the actin filament.**
 - D. the actin head attaches to the myosin filament.
 - E. the myosin head initiates the power stroke.
-

31. The fluid that moves around in the circulatory system of a typical arthropod is the
- A. blood plasma.
 - B. hemolymph.**
 - C. cytoplasm.
 - D. gastrovascular fluid.
 - E. intracellular fluid.
32. The highest quality form of energy (best able to do work) for organisms is
- A. kinetic.
 - B. radiant.**
 - C. potential.
 - D. thermal.
 - E. chemical.
33. During protist conjugation, which of the following does *not* occur during some part of the process?
- A. The micronucleus undergoes mitosis.
 - B. Some micronuclei disintegrate.
 - C. The macronucleus disintegrates.
 - D. The micronucleus undergoes meiosis.
 - E. Cells exchange micronuclei.
 - F. The macronucleus fuses with the micronucleus.**
34. The ion used most often as a second messenger is
- A. Ca^{+}**
 - B. K^{+}
 - C. Cl^{-}
 - D. Na^{+}
 - E. H^{+}
35. The main stimulus used by birds to find their way north for the summer time is
- A. electromagnetic.**
 - B. mechanical.
 - C. thermal.
 - D. chemical.
 - E. thigmic.
36. Transpiration in plants requires all of the following *except*
- A. adhesion of water molecules to cellulose.
 - B. evaporation of water molecules.
 - C. cohesion between water molecules.
 - D. water vapor on the spongy mesophyll.
 - E. active transport through xylem cells.**
-

37. Differences in energy quality and availability affects the amount of surface area and volume in organisms. Compared to animals, plants have
- A. **high surface area and low volumes.**
 - B. low surface area and high volumes.
 - C. low surface areas and low volumes.
 - D. high surface area and high volumes.
38. The function of valves in veins in a closed circulatory system is to
- A. provide support for the extra muscles around the vein.
 - B. decrease the amount of pressure in the veins.
 - C. **ensure one way flow of circulatory fluids.**
 - D. enable the vein to withstand high pressure.
 - E. enhance diffusion of nutrients to the tissues.
39. Pores on the leaf surface that function in gas and water exchange are called
- A. phloem.
 - B. xylem.
 - C. waxy cuticles.
 - D. spongy mesophyll.
 - E. **stomata.**
40. Hydrostatic skeletons are
- A. external skeletons that provide support and protection against dehydration.
 - B. **fluid-filled compartments under pressure.**
 - C. internal skeletons that are buried in soft tissue.
 - D. skeletons made of microtubules in the cytoplasm of protists.
 - E. skeletons found in deep sea organisms that depend on high pressure of the surrounding water to provide support.
41. What drives the flow of water through the xylem?
- A. Transpirational pull through sieve-tube elements.
 - B. Absciscic acid opening guard cells around the stomata.
 - C. A higher concentration of sugars in the leaves than in the roots.
 - D. **The evaporation of water from the leaves.**
 - E. The number of companion cells in the phloem.
42. The type of nerve cell responsible for integrating different stimuli is the
- A. motor neuron
 - B. axon
 - C. dendrite
 - D. **interneuron**
 - E. sensory neuron
-

43. The protein that is activated in response to auxin that allows phototropism to occur is
- A. expansin.**
 - B. proton channel.
 - C. wallulase.
 - D. transmembrane.
 - E. abscisic acid.
 - F. basal apparatus.
44. Organisms with a circulating fluid that is separate from the fluid that directly surrounding the body's cells are likely to have
- A. a vascular system.
 - B. a hemolymph system.
 - C. a gastrovascular cavity.
 - D. a closed circulatory system.**
 - E. an open circulatory system.
45. From an energy perspective, animals are mobile because they consume
- A. high quality energy and carbon intake is irregular.
 - B. low quality energy and carbon intake is constant.
 - C. high quality energy and carbon intake is constant.
 - D. low quality energy and low energy efficiency.
 - E. low quality energy and carbon intake is irregular.**
 - F. high quality energy and high energy efficiency.
46. The vascular tissue system for delivery of water, minerals, and nutrients throughout the plant includes all of the following except for
- A. cohesion.**
 - B. sieve plates.
 - C. phloem.
 - D. companion cells.
 - E. xylem.
47. Ignoring all other factors, what kind of day would result in the fastest delivery of water and minerals to the leaves of a tree?
- A. A warm, dry day.**
 - B. A cool, humid day.
 - C. A cool, dry day.
 - D. A very hot, dry, windy day.
 - E. A warm, humid day.
-

48. Which of the following are primarily responsible for movement in the eukaryotic flagellum?

- A. Actin and myosin filaments.
- B. Hydrogen ions and basal apparatus.
- C. Dynein and microtubules.**
- D. Basal apparatus and hook.
- E. Sensory and motor neurons.

49. Which of the following about phloem is false?

- A. The plasma membrane is modified to form a sieve plate.**
- B. Sugars move by passive diffusion.
- C. The cell has cytoplasm.
- D. Companion cells provide proteins to sieve-tube elements.
- E. Sieve-tube elements do not have a nucleus.

Because they have to divide to multiply. Be wise, do good things, have a great summer!
